



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ALAMU AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

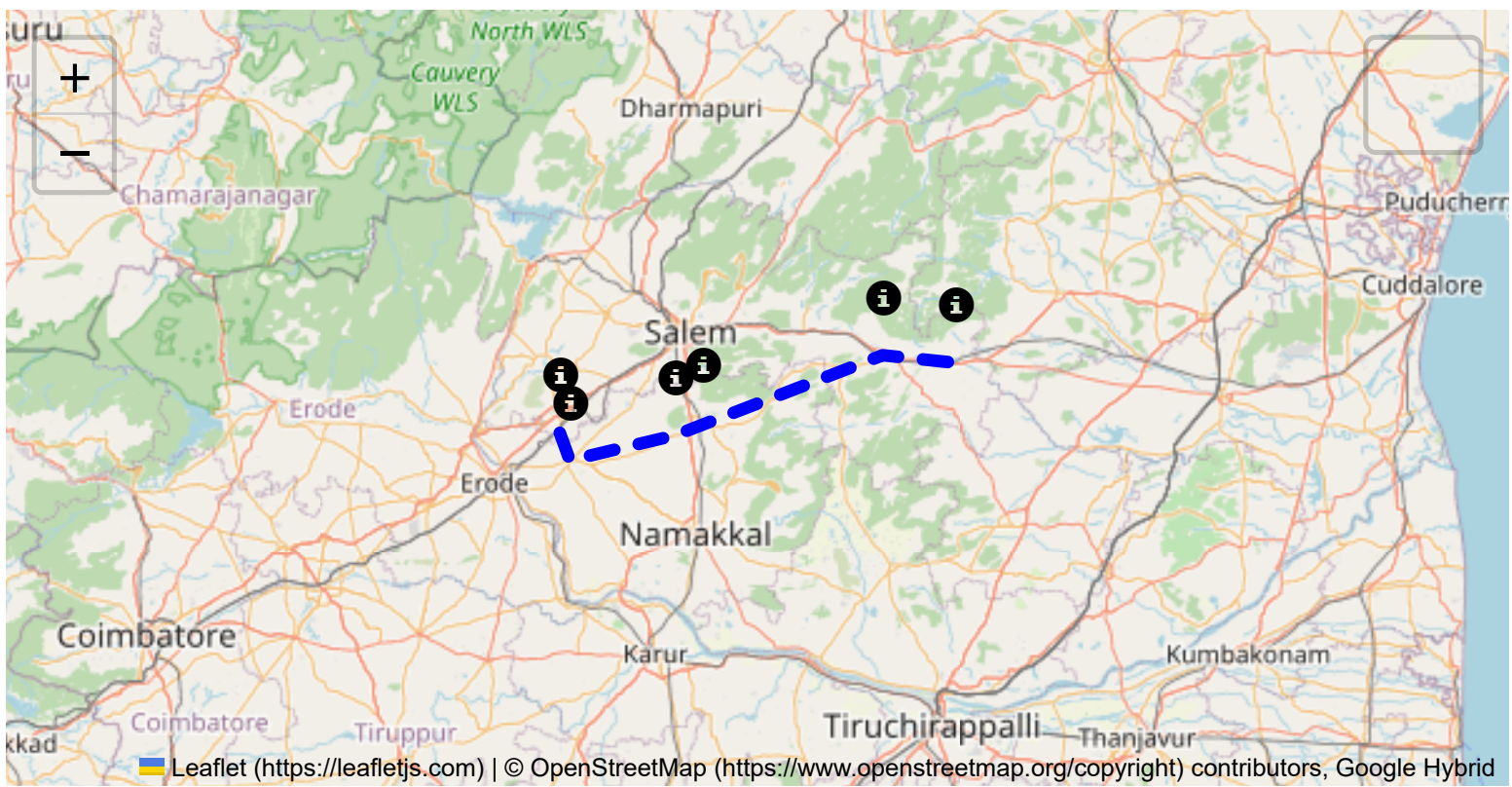
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 116.58 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours
Start: (11.4381, 77.8734)
End: (11.585862, 78.736466)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to the Salem - Ulundurpettai Highway spans approximately 116.58 kilometers with a typical travel time of around 2.59 hours for heavy vehicles. The journey covers several major waypoints like Kamalar, Ponkurichi, Rasipuram, and Narasingapuram. This route utilizes state highways and local roads, traversing both rural and semi-urban areas.

Weather Conditions and Hazards

- **Typical Weather:** The region experiences a tropical climate with hot summers, mild winters, and substantial rainfall during the monsoon. Temperature variations can affect road conditions, especially during peak summer months.
- **Weather-Related Hazards:** Flash floods during the monsoon can cause waterlogging, while heat can lead to road material expansion. Sudden weather changes can also lead to reduced visibility.

Traffic Patterns

- **Peak Hours:** Busy traffic periods typically occur in the early mornings (7-9 AM) and late afternoons (5-8 PM), with congestion in Tiruchengode and Rasipuram being particularly noticeable.
- **Congestion-Prone Areas:** Rasipuram and Narasingapuram, being closer to NH 79, often experience higher traffic volumes due to their connectivity to smaller townships.

Road Quality and Infrastructure

- **State Highways** (e.g., SH 79): Generally well-maintained with appropriate signage and wider lanes, but occasional potholes and uneven surfaces exist.
- **Local Roads:** These can be narrower and less well-maintained, requiring caution. Roadside infrastructure such as medians and guard rails may be inconsistent.

Emergency Alternative Routes

In emergencies, alternate routes via larger highways like NH 44 can provide more reliable road conditions and quicker access to emergency services. However, traffic conditions should be evaluated in real-time.

Local Regulations

- **Hazardous Material Transport:** There are stringent restrictions on the type of materials that can be transported. Drivers should be well-versed in local regulations and ensure proper documentation and safety measures are in place.

Historical Incidents

- **Heavy Vehicle Incidents:** While there are occasional reports of incidents involving heavy vehicles in Rasipuram, major accidents with hazardous materials have been relatively rare. However, minor spills and collisions due to negligence highlight the need for vigilance.

Environmental Considerations

- **Sensitive Areas:** The route crosses agricultural zones, requiring careful adherence to environmental safety protocols to prevent chemical spills or pollution.
- **Wildlife:** Although there are no officially protected forests directly on the route, it's important to be aware of crossing animals particularly during night time.

Communication Coverage

- Potential Dead Zones:** Cellular coverage along the main highways is generally consistent, but rural stretches and portions near Ponkurichi may experience weak signals.

Emergency Response Times

- Response Times:** Proximity to larger towns like Rasipuram and Tiruchengode implies faster response times, typically within 20-30 minutes. Rural areas might face delays up to an hour due to terrain and road conditions.

Overall Risk Assessment Summary

The route is feasible for heavy vehicle travel but demands careful attention to road quality variations and weather-induced hazards. While the infrastructure is supportive in most parts, it's paramount to stay informed about real-time traffic updates and weather conditions to mitigate risks. Regular communication checks and environmental compliance are vital, alongside maintaining awareness of local regulations and ensuring clear emergency strategies are in place.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.37902, 77.89500	30 KM/Hr
5	Turn	High	11.37862, 77.89524	15 KM/Hr
6	Turn	Medium	11.37809, 77.89834	30 KM/Hr
7	Turn	Medium	11.37885, 77.89989	30 KM/Hr
8	Turn	Medium	11.37820, 77.90220	30 KM/Hr
9	Turn	Medium	11.36748, 77.93938	30 KM/Hr
10	Turn	Medium	11.38968, 78.02925	30 KM/Hr
11	Turn	Medium	11.41288, 78.08045	30 KM/Hr
12	Turn	High	11.41466, 78.08070	15 KM/Hr
13	Turn	High	11.43253, 78.12084	15 KM/Hr
14	Turn	High	11.45587, 78.18054	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	Medium	11.45757, 78.18137	30 KM/Hr
16	Blind Spot	Blind Spot	11.46032, 78.18514	10 KM/Hr
17	Turn	High	11.45898, 78.18564	15 KM/Hr
18	Turn	Medium	11.44993, 78.22152	30 KM/Hr
19	Turn	High	11.44951, 78.22263	15 KM/Hr
20	Turn	High	11.44797, 78.22314	15 KM/Hr
21	Turn	Medium	11.44756, 78.22414	30 KM/Hr
22	Turn	Medium	11.45353, 78.23584	30 KM/Hr
23	Turn	Medium	11.49835, 78.32417	30 KM/Hr
24	Turn	Medium	11.50865, 78.33966	30 KM/Hr
25	Turn	Medium	11.51183, 78.34416	30 KM/Hr
26	Turn	Medium	11.51870, 78.34816	30 KM/Hr
27	Turn	Medium	11.54842, 78.35627	30 KM/Hr
28	Turn	Medium	11.56719, 78.37813	30 KM/Hr
29	Turn	Medium	11.56940, 78.39856	30 KM/Hr
30	Turn	Medium	11.58180, 78.41812	30 KM/Hr
31	Turn	Medium	11.57108, 78.49917	30 KM/Hr
32	Turn	Medium	11.57105, 78.50084	30 KM/Hr
33	Turn	Medium	11.57829, 78.51245	30 KM/Hr
34	Turn	Medium	11.58681, 78.54334	30 KM/Hr
35	Turn	Medium	11.58679, 78.54553	30 KM/Hr
36	Turn	Medium	11.59003, 78.55130	30 KM/Hr
37	Turn	Medium	11.59108, 78.55637	30 KM/Hr
38	Turn	Medium	11.59253, 78.55778	30 KM/Hr
39	Turn	High	11.59744, 78.57347	15 KM/Hr
40	Turn	High	11.59553, 78.57461	15 KM/Hr
41	Turn	High	11.59538, 78.57598	15 KM/Hr
42	Turn	High	11.59861, 78.57703	15 KM/Hr
43	Blind Spot	Blind Spot	11.60109, 78.58219	10 KM/Hr
44	Turn	Medium	11.60434, 78.57636	30 KM/Hr
45	Turn	High	11.60491, 78.57377	15 KM/Hr
46	Turn	Medium	11.61294, 78.57255	30 KM/Hr
47	Turn	High	11.61426, 78.57274	15 KM/Hr
48	Turn	Medium	11.59340, 78.62042	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
15	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
16	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
17	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
18	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
19	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
20	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
21	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
22	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium
23	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium
24	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
26	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium
28	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
29	clinic	Dr. Saravanakumar Clinic	11.5842002, 78.4441223	30 km/h	Medium
30	hospital	Government Hospital	11.572472, 78.497407	30 km/h	Medium
32	hospital	Abi Hospital	11.59319, 78.560129	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
9	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
10	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
11	school	Vidya madir school	11.4356335, 78.1381408	30 km/h	Medium
12	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
13	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
14	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium
25	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium
31	school	Rasi Matric HR Sec School	11.5760027, 78.5062808	30 km/h	Medium
33	school	Saraswati Metric Hr sec School	11.5910827, 78.6402264	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
34	school	MARUTHI PRIMAR, SECONDAR ,HIGHER SECONDARY SHOOL	11.5859471, 78.7186322	30 km/h	Medium
35	college	Maruthi Polytechnic College	11.5856852, 78.7201072	30 km/h	Medium
36	college	Bharathiyar Educational Institure	11.5863463, 78.7364793	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37902, 77.89500



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37809, 77.89834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37885, 77.89989



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37820, 77.90220



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.36748, 77.93938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38968, 78.02925



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41288, 78.08045



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41466, 78.08070

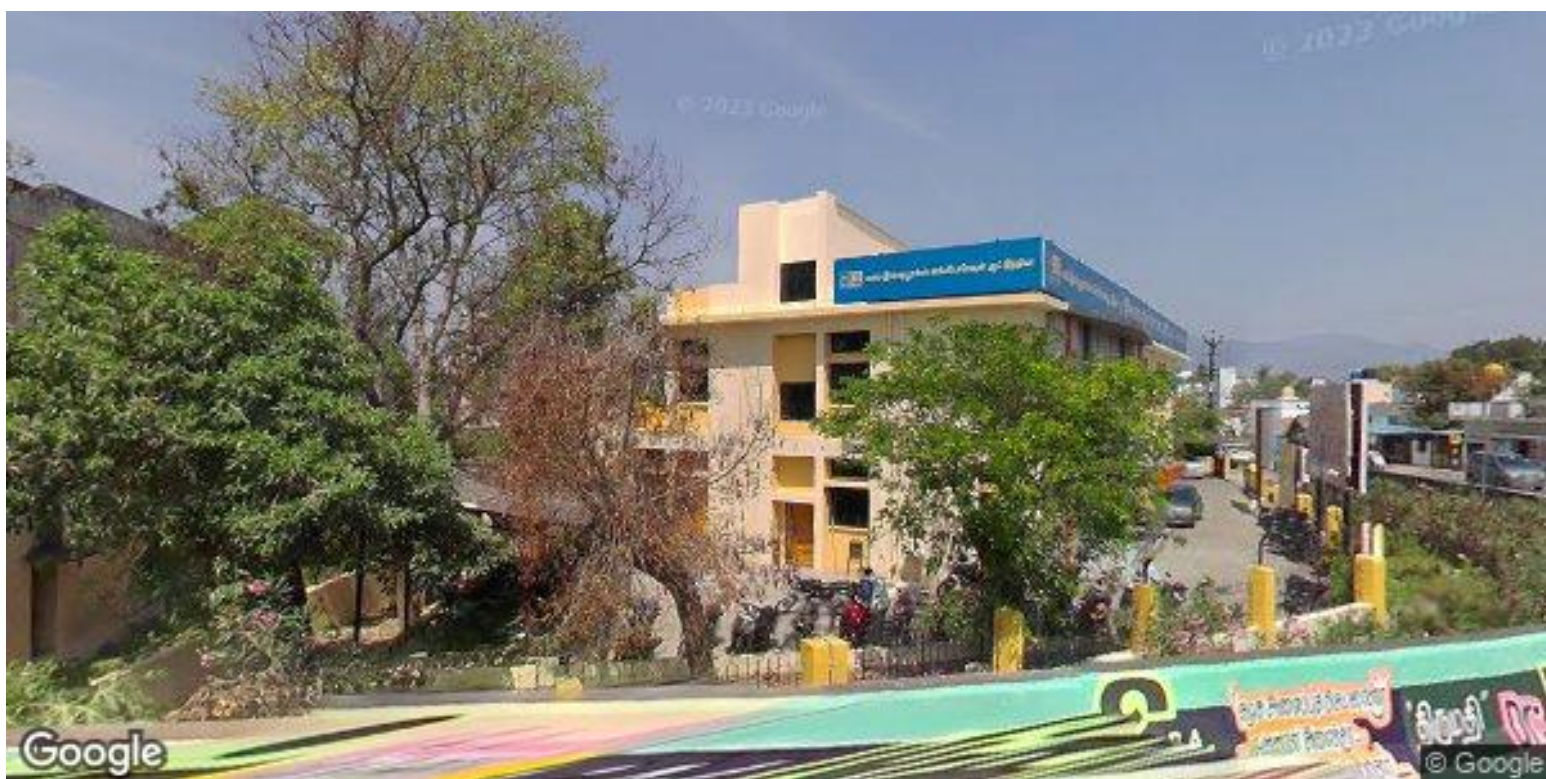


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43253, 78.12084



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45587, 78.18054



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45757, 78.18137



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.46032, 78.18514



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45898, 78.18564



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44993, 78.22152



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44951, 78.22263



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44797, 78.22314



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44756, 78.22414



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 78.23584



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49835, 78.32417

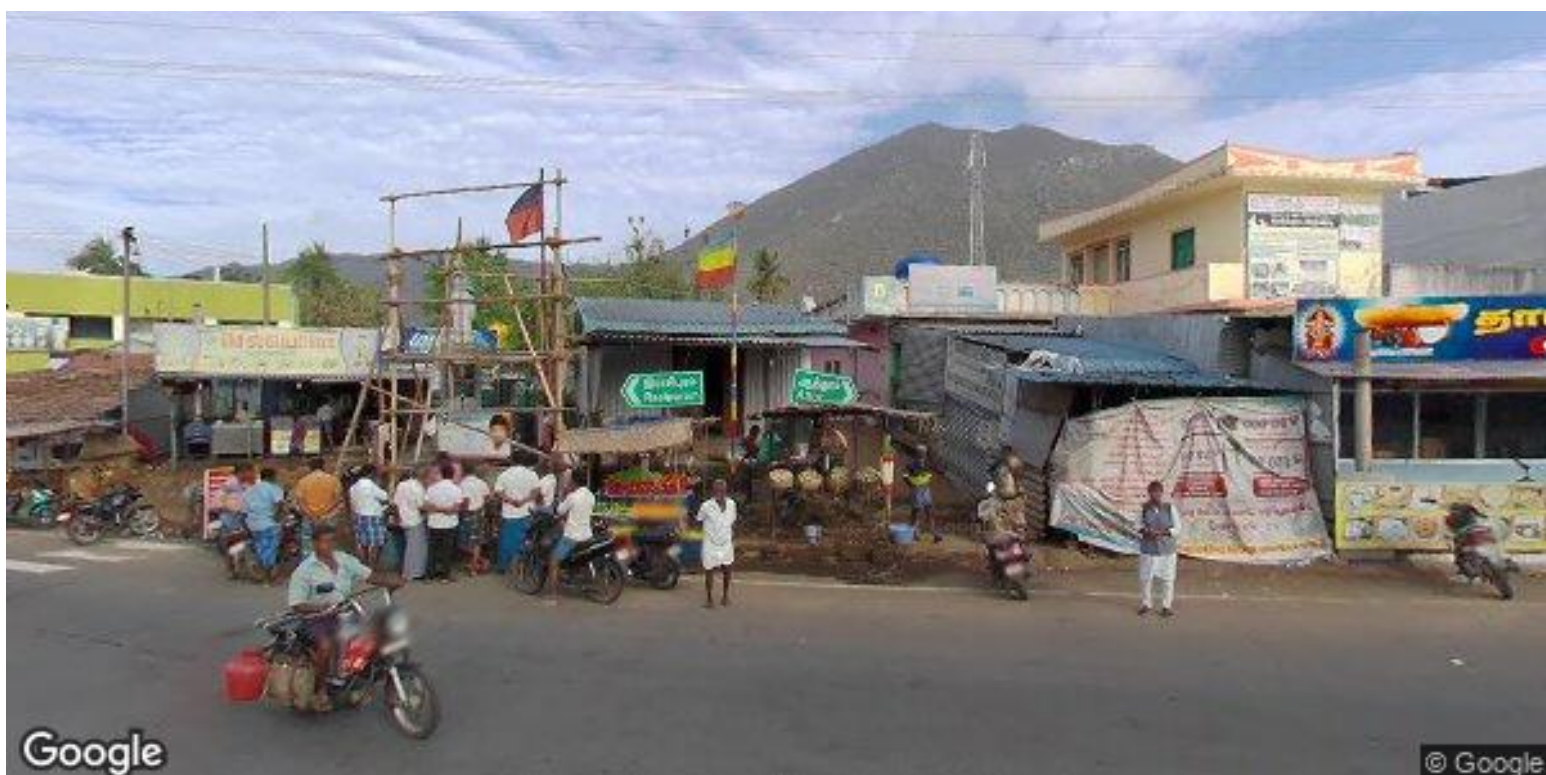


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50865, 78.33966



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51183, 78.34416



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51870, 78.34816



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54842, 78.35627



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56719, 78.37813



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56940, 78.39856



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58180, 78.41812



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57108, 78.49917



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57105, 78.50084



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57829, 78.51245



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58679, 78.54553



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59744, 78.57347



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59553, 78.57461



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59538, 78.57598



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59861, 78.57703



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.60109, 78.58219



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60434, 78.57636



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60491, 78.57377



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61294, 78.57255



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61426, 78.57274



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59340, 78.62042
